

TutorHub - Complete Requirements Document

Document Information

Property	Value
Project Name	TutorHub - Online Tutoring Platform
Document Type	Software Requirements Specification (SRS)
Version	1.0
Date	2026-04-19
Target Platform	Web Application (ASP.NET Core)

Table of Contents

1. Project Overview
 2. User Roles
 3. Functional Requirements by Module
 4. Non-Functional Requirements
 5. Business Rules
 6. Integration Requirements
 7. Security Requirements
 8. Reporting Requirements
-

1. Project Overview

1.1 Vision

TutorHub is an online platform connecting students/parents with qualified tutors for personalized learning sessions. The platform enables tutor discovery, booking, payments, and real-time communication.

1.2 Core Objectives

- Provide verified, high-quality tutors across multiple subjects
- Enable seamless booking and payment process
- Facilitate real-time communication between tutors and students
- Ensure secure document verification and identity management
- Provide analytics and earnings dashboard for tutors

1.3 Scope

- Web-based application accessible via browsers
 - Multi-role system (Student, Parent, Tutor, Admin)
 - Real-time chat functionality
 - Payment processing integration
 - Document management and verification
 - Search with filters and recommendations
-

2. User Roles

2.1 Role Matrix

Role	Description	Access Level
Student	Individual seeking tutoring	Can search, book, pay, review
Parent	Parent managing child's tutoring	Can manage multiple children, book, track progress
Tutor	Educator providing teaching services	Can manage profile, availability, earnings
Admin	Platform administrator	Full system access, user management, approvals

2.2 Role Capabilities Matrix

Feature	Student	Parent	Tutor	Admin
---------	---------	--------	-------	-------

View Tutor Profiles	✓	✓	✓	✓
Search/Filter Tutors	✓	✓	-	✓
Book Sessions	✓	✓	-	-
Make Payments	✓	✓	-	-
View Bookings	✓	✓	✓	✓
Cancel/Reschedule	✓	✓	✓	-
Rate & Review	✓	✓	-	-
Create/Edit Profile	✓	✓	✓	-
Set Availability	-	-	✓	-
Upload Documents	-	-	✓	-
View Earnings	-	-	✓	✓
Request Payout	-	-	✓	-
Chat with Users	✓	✓	✓	✓
Manage Users	-	-	-	✓
Approve Tutors	-	-	-	✓
View Analytics	-	-	-	✓
Manage Platform Fees	-	-	-	✓

3. Functional Requirements by Module

3.1 Authentication & Account Management

REG-01: User Registration

ID	Requirement	Priority
REG-01-01	User can register as Student, Parent, or Tutor	High
REG-01-02	Registration requires: First Name, Last Name, Email, Password, Role	High
REG-01-03	Email verification required before first login	High

REG-01-04	Password must meet strength requirements (min 8 chars, 1 uppercase, 1 number, 1 special)	High
REG-01-05	Duplicate email addresses not allowed	High
REG-01-06	Parent can link multiple student profiles after registration	Medium

REG-02: User Login

ID	Requirement	Priority
REG-02-01	User can login with email and password	High
REG-02-02	JWT token returned on successful authentication	High
REG-02-03	Refresh token mechanism for session persistence	High
REG-02-04	"Remember Me" option for 30-day session	Medium
REG-02-05	Max 5 failed login attempts before 15-minute lockout	High

REG-03: Password Management

ID	Requirement	Priority
REG-03-01	User can request password reset via email	High
REG-03-02	Password reset link expires in 1 hour	High
REG-03-03	User can change password while logged in	High
REG-03-04	Two-Factor Authentication (2FA) via email or SMS	Medium

REG-04: User Profile Management

ID	Requirement	Priority
REG-04-01	User can view and edit profile information	High
REG-04-02	User can upload profile picture	Medium
REG-04-03	User can update email (requires verification)	Medium
REG-04-04	User can deactivate account	Low

3.2 Tutor Management Module

TUT-01: Tutor Profile Creation

ID	Requirement	Priority
TUT-01-01	Tutor must complete profile after registration	High
TUT-01-02	Profile includes: Headline, Bio, Education, Experience, Location	High
TUT-01-03	Tutor selects subjects they teach (multiple selection)	High
TUT-01-04	Tutor selects grade levels they cover	High
TUT-01-05	Tutor sets hourly rate (min and max)	High
TUT-01-06	Tutor selects teaching mode (Online, In-Person, Both)	High

TUT-02: Document Verification

ID	Requirement	Priority
TUT-02-01	Tutor must upload government ID for verification	High
TUT-02-02	Tutor must upload education certificates	High
TUT-02-03	Tutor can upload teaching certifications (optional)	Medium
TUT-02-04	Admin reviews and approves/rejects documents	High
TUT-02-05	Verified badge appears on tutor profile after approval	High
TUT-02-06	Tutor notified via email of verification status	High

TUT-03: Availability Management

ID	Requirement	Priority
TUT-03-01	Tutor can set weekly recurring availability	High
TUT-03-02	Tutor can set specific date exceptions	Medium
TUT-03-03	Tutor can block off time for holidays/vacation	Medium
TUT-03-04	System prevents double-booking of same time slot	High
TUT-03-05	Tutor can view their schedule calendar	High
TUT-03-06	Availability automatically updates when booking confirmed	High

TUT-04: Tutor Dashboard

ID	Requirement	Priority
TUT-04-01	View upcoming sessions list	High
TUT-04-02	View past sessions history	High
TUT-04-03	View earnings summary (daily/weekly/monthly/yearly)	High
TUT-04-04	View student reviews and ratings	High
TUT-04-05	View profile views and booking conversion stats	Medium
TUT-04-06	Download earnings report (CSV/PDF)	Medium

TUT-05: Tutor Earnings & Payouts

ID	Requirement	Priority
TUT-05-01	View total earnings (including platform fee breakdown)	High
TUT-05-02	Request payout to connected bank account	High
TUT-05-03	View payout history and status	High
TUT-05-04	Minimum \$50 for payout request	Medium
TUT-05-05	Automatic weekly payouts option (opt-in)	Low

3.3 Student/Parent Module

STU-01: Tutor Discovery

ID	Requirement	Priority
STU-01-01	Search tutors by subject, name, or keyword	High
STU-01-02	Filter by price range	High
STU-01-03	Filter by rating (1-5 stars)	High
STU-01-04	Filter by teaching mode (Online/In-Person)	High
STU-01-05	Filter by location/city	High
STU-01-06	Filter by grade level	High
STU-01-07	Sort by rating, price, experience, relevance	High

STU-01-08	View tutor profile with complete details	High
-----------	--	------

STU-02: Booking Process

ID	Requirement	Priority
STU-02-01	Select tutor, subject, date, time slot	High
STU-02-02	Choose duration (30, 60, 90, 120 minutes)	High
STU-02-03	Select teaching mode (Online/In-Person)	High
STU-02-04	Add notes/special requests for tutor	Medium
STU-02-05	View price calculation before booking	High
STU-02-06	Send booking request to tutor	High
STU-02-07	Receive notification when tutor responds	High

STU-03: Booking Management

ID	Requirement	Priority
STU-03-01	View all bookings (upcoming, past, cancelled)	High
STU-03-02	Cancel booking (free up to 24 hours before)	High
STU-03-03	Request reschedule (requires tutor approval)	High
STU-03-04	Join online session via meeting link	High
STU-03-05	Mark session as completed	Medium
STU-03-06	Report issue with session	Medium

STU-04: Payment Processing

ID	Requirement	Priority
STU-04-01	Pay for booking using credit/debit card	High
STU-04-02	View payment breakdown (tutor fee + platform fee)	High
STU-04-03	Receive payment confirmation receipt	High
STU-04-04	View payment history	High
STU-04-05	Request refund (subject to cancellation policy)	Medium
STU-04-06	Save payment method for future bookings	Medium

STU-05: Reviews & Ratings

ID	Requirement	Priority
STU-05-01	Rate tutor (1-5 stars) after completed session	High
STU-05-02	Write detailed review comment	High
STU-05-03	Select tags (e.g., "Clear explanations", "Patient")	Medium
STU-05-04	Edit/delete review within 7 days	Low
STU-05-05	View all reviews for a tutor	High

3.4 Parent-Specific Module

PAR-01: Multiple Student Management

ID	Requirement	Priority
PAR-01-01	Parent can create profiles for multiple children	High
PAR-01-02	Each child profile has name, grade level, subjects	High
PAR-01-03	Book sessions on behalf of any child	High
PAR-01-04	View progress reports per child	Medium
PAR-01-05	Receive session summaries for each child	Medium

PAR-02: Parent Dashboard

ID	Requirement	Priority
PAR-02-01	Overview of all children's upcoming sessions	High
PAR-02-02	View total spending across all children	High
PAR-02-03	Track child's improvement over time	Medium
PAR-02-04	Receive notifications for each child's sessions	High

3.5 Chat & Communication Module

CHT-01: Real-Time Chat

ID	Requirement	Priority
CHT-01-01	Student and tutor can chat before/during/after sessions	High
CHT-01-02	Real-time message delivery using SignalR	High
CHT-01-03	Message read receipts (seen/delivered status)	High
CHT-01-04	Typing indicators	Medium
CHT-01-05	Online/offline status display	Medium
CHT-01-06	File sharing (images, documents) in chat	Medium

CHT-02: Chat History

ID	Requirement	Priority
CHT-02-01	Persistent chat history stored in database	High
CHT-02-02	Search chat history by keyword	Low
CHT-02-03	Export chat conversation	Low

3.6 Admin Module

ADM-01: User Management

ID	Requirement	Priority
ADM-01-01	View all users with filters (role, status, date joined)	High
ADM-01-02	Suspend/activate user accounts	High
ADM-01-03	Delete user accounts (GDPR compliance)	High
ADM-01-04	Change user role	High
ADM-01-05	Reset user password	Medium

ADM-02: Tutor Approval Workflow

ID	Requirement	Priority
ADM-02-01	View pending tutor applications	High
ADM-02-02	Review uploaded documents (ID, certificates)	High
ADM-02-03	Approve tutor with optional notes	High
ADM-02-04	Reject tutor with reason	High

ADM-02-05	Request additional documents from tutor	Medium
ADM-02-06	Audit log of all approval actions	Medium

ADM-03: Booking Management

ID	Requirement	Priority
ADM-03-01	View all platform bookings	High
ADM-03-02	Cancel any booking (with reason)	High
ADM-03-03	Process refunds manually	High
ADM-03-04	View dispute resolution cases	Medium

ADM-04: Financial Management

ID	Requirement	Priority
ADM-04-01	View platform revenue (daily/weekly/monthly)	High
ADM-04-02	Configure platform fee percentage	High
ADM-04-03	View all payouts to tutors	High
ADM-04-04	Generate financial reports	Medium
ADM-04-05	Export transaction history (CSV/Excel)	Medium

ADM-05: Platform Analytics

ID	Requirement	Priority
ADM-05-01	Dashboard with key metrics (users, bookings, revenue)	High
ADM-05-02	Chart: Bookings over time (daily/weekly/monthly)	High
ADM-05-03	Chart: Top subjects by booking volume	High
ADM-05-04	Chart: Top performing tutors	High
ADM-05-05	User growth analytics	Medium
ADM-05-06	Retention and churn rates	Low

ADM-06: Content Management

ID	Requirement	Priority
ADM-06-01	Add/edit/delete subjects	High

ADM-06-02	Add/edit/delete grade levels	High
ADM-06-03	Manage system notification messages	Medium
ADM-06-04	Manage FAQ content	Low

3.7 Notification Module

NTF-01: Notification Types

ID	Notification Type	Trigger	Channel
NTF-01-01	Welcome Email	After registration	Email
NTF-01-02	Email Verification	After registration	Email
NTF-01-03	Booking Request Received	Student requests booking	Email + In-app
NTF-01-04	Booking Confirmed	Tutor accepts booking	Email + SMS + In-app
NTF-01-05	Booking Cancelled	Either party cancels	Email + In-app
NTF-01-06	Booking Reminder	1 hour before session	Email + SMS
NTF-01-07	Payment Confirmation	After successful payment	Email + In-app
NTF-01-08	New Message	Chat message received	In-app + Email
NTF-01-09	Tutor Approval	Admin approves tutor	Email
NTF-01-10	Payout Processed	After payout sent	Email
NTF-01-11	Review Received	Student submits review	Email + In-app

NTF-01-12	Session Rescheduled	After reschedule	Email + In-app
-----------	---------------------	------------------	----------------

NTF-02: Notification Preferences

ID	Requirement	Priority
NTF-02-01	User can enable/disable email notifications	Medium
NTF-02-02	User can enable/disable SMS notifications	Medium
NTF-02-03	User can set quiet hours	Low

3.8 Payment & Billing Module

PAY-01: Payment Processing

ID	Requirement	Priority
PAY-01-01	Accept credit/debit card payments (Visa, Mastercard, Amex)	High
PAY-01-02	Secure payment processing via Stripe	High
PAY-01-03	Calculate platform fee (configurable percentage)	High
PAY-01-04	Calculate tutor earnings (amount - platform fee)	High
PAY-01-05	Support multiple currencies	Low

PAY-02: Refund Policy

ID	Requirement	Priority
PAY-02-01	Full refund if cancelled 24+ hours before session	High
PAY-02-02	50% refund if cancelled 12-24 hours before session	High
PAY-02-03	No refund if cancelled less than 12 hours before	High
PAY-02-04	Admin can override refund policy	Medium

PAY-03: Tutor Payouts

ID	Requirement	Priority
PAY-03-01	Tutor must connect Stripe Connect account	High
PAY-03-02	Payouts processed weekly (every Monday)	High
PAY-03-03	Minimum payout threshold: \$50	High
PAY-03-04	Payout includes completed sessions from previous week	High
PAY-03-05	Payout statement shows per-session breakdown	Medium

3.9 Search Module

SRH-01: Search Functionality

ID	Requirement	Priority
SRH-01-01	Full-text search by tutor name, subject, keyword	High
SRH-01-02	Autocomplete suggestions while typing	Medium
SRH-01-03	Spell correction for search terms	Low
SRH-01-04	Search results show tutor card with: name, rating, price, subjects, verified badge	High
SRH-01-05	Pagination (20 results per page)	High

SRH-02: Advanced Filtering

ID	Requirement	Priority
SRH-02-01	Price slider (min-max)	High
SRH-02-02	Rating filter (4+, 4.5+, 5)	High
SRH-02-03	Subject multi-select	High
SRH-02-04	Grade level multi-select	High
SRH-02-05	Teaching mode (Online/In-Person/Both)	High
SRH-02-06	Availability filter (Today, Tomorrow, This Week)	Medium
SRH-02-07	Language filter	Low

SRH-03: Search Ranking

ID	Requirement	Priority
SRH-03-01	Tutors sorted by relevance (search term matching)	High

SRH-03-02	Higher rating tutors rank better	High
SRH-03-03	Verified tutors get ranking boost	High
SRH-03-04	Recently active tutors get ranking boost	Medium
SRH-03-05	Featured tutors (paid placement)	Low

4. Non-Functional Requirements

4.1 Performance Requirements

ID	Requirement	Target
NFR-01	Page load time	< 3 seconds
NFR-02	Search response time	< 500 ms
NFR-03	API response time (95th percentile)	< 200 ms
NFR-04	Chat message delivery latency	< 1 second
NFR-05	Concurrent users supported	10,000
NFR-06	Database query optimization	All queries < 100 ms

4.2 Security Requirements

ID	Requirement	Priority
NFR-07	All passwords hashed using bcrypt	High
NFR-08	HTTPS enforced for all communications	High
NFR-09	JWT tokens expire after 60 minutes	High
NFR-10	Refresh token rotation implemented	High
NFR-11	Rate limiting (100 requests per minute per IP)	High
NFR-12	SQL injection prevention using EF Core	High
NFR-13	XSS prevention through output encoding	High
NFR-14	CSRF protection enabled	High
NFR-15	API request logging for audit	Medium
NFR-16	GDPR compliance (data export, delete)	High

4.3 Availability Requirements

ID	Requirement	Target
NFR-17	System uptime	99.9%
NFR-18	Maintenance window notification	48 hours advance
NFR-19	Database backup frequency	Daily
NFR-20	Disaster recovery time	< 4 hours

4.4 Scalability Requirements

ID	Requirement	Priority
NFR-21	Horizontal scaling support for API	Medium
NFR-22	Database read replicas for reporting	Low
NFR-23	Redis caching for frequent queries	Medium
NFR-24	Elasticsearch cluster for search scaling	Medium

4.5 Usability Requirements

ID	Requirement	Priority
NFR-25	Mobile responsive design	High
NFR-26	WCAG 2.1 AA accessibility compliance	Medium
NFR-27	Support for screen readers	Medium
NFR-28	Keyboard navigation support	Low

4.6 Compliance Requirements

ID	Requirement	Priority
NFR-29	PCI DSS compliance for payments (via Stripe)	High
NFR-30	COPPA compliance (under 13 data collection)	Medium
NFR-31	Data retention policy (3 years)	Medium

5. Business Rules

5.1 Booking Rules

ID	Rule
BR-01	Minimum booking duration: 30 minutes
BR-02	Maximum booking duration: 4 hours
BR-03	Booking must be made at least 2 hours before start time
BR-04	Maximum future booking: 60 days in advance
BR-05	Tutor must respond to booking request within 24 hours
BR-06	Unanswered booking requests auto-cancel after 24 hours

5.2 Cancellation Rules

ID	Rule
BR-07	Free cancellation: 24+ hours before session
BR-08	50% refund: 12-24 hours before session
BR-09	No refund: less than 12 hours before session
BR-10	Tutor cancellation: student receives full refund
BR-11	No-show student: no refund, tutor paid 50%
BR-12	No-show tutor: full refund + \$10 credit to student

5.3 Platform Fee Rules

ID	Rule
BR-13	Platform fee: 10% of session amount
BR-14	Fee applies to all completed sessions
BR-15	Fee waived for first session (new student promotion)

5.4 Verification Rules

ID	Rule
BR-16	Tutors must be 18+ years old
BR-17	Government ID required for verification
BR-18	Education certificate required for each subject taught

BR-19	Verification takes 1-3 business days
-------	--------------------------------------

5.5 Review Rules

ID	Rule
BR-20	Review can only be submitted after session completion
BR-21	Review window: 14 days after session
BR-22	One review per booking
BR-23	Admin can remove reviews violating guidelines
BR-24	Tutors cannot review students

6. Integration Requirements

6.1 External Services

Service	Purpose	Priority
Stripe	Payment processing, payouts	High
SendGrid	Email notifications	High
Twilio	SMS notifications	Medium
Azure Blob Storage	Document and image storage	High
Elasticsearch	Full-text search	Medium
Redis	Caching	Medium

6.2 API Endpoints Summary

Module	Base Route	# Endpoints
Authentication	/api/auth	8
Tutor Management	/api/tutor	15
Student/Parent	/api/student	12
Booking	/api/booking	12

Payment	/api/payment	8
Review	/api/review	6
Chat	/api/chat	5
Search	/api/search	4
Notification	/api/notification	6
Admin	/api/admin	18
Total		94

7. Security Requirements

7.1 Authentication & Authorization

ID	Requirement
SEC-01	JWT tokens with 60-minute expiration
SEC-02	Refresh tokens valid for 7 days
SEC-03	Role-based access control (RBAC)
SEC-04	Password reset tokens expire in 1 hour
SEC-05	Email verification required before login

7.2 Data Protection

ID	Requirement
SEC-06	Passwords hashed using bcrypt (cost factor 12)
SEC-07	Sensitive data (payment info) never stored
SEC-08	All personal data encrypted at rest
SEC-09	Audit logs for all admin actions

7.3 API Security

ID	Requirement
SEC-10	Rate limiting: 100 requests/minute per IP
SEC-11	CORS configured for trusted domains only
SEC-12	Request validation on all endpoints
SEC-13	API keys for external service calls

8. Reporting Requirements

8.1 Tutor Reports

ID	Report Name	Frequency
RPT-01	Earnings Summary	Real-time
RPT-02	Session History	Real-time
RPT-03	Student Feedback Report	Monthly
RPT-04	Payout Statement	Weekly

8.2 Admin Reports

ID	Report Name	Frequency
RPT-05	Platform Revenue Report	Daily/Weekly/Monthly
RPT-06	User Growth Report	Monthly
RPT-07	Top Tutors Report	Monthly
RPT-08	Popular Subjects Report	Monthly
RPT-09	Cancellation Analysis	Monthly
RPT-10	Dispute Resolution Report	Monthly

8.3 Export Formats

- CSV

- Excel (XLSX)
 - PDF
-

9. Success Criteria

9.1 Launch Requirements (MVP)

- User registration and login (Student, Tutor, Admin)
- Tutor profile creation with subjects and rates
- Basic tutor search with filters
- Booking request and confirmation flow
- Payment processing via Stripe
- Basic admin dashboard for tutor approvals
- Email notifications

9.2 Phase 2 Requirements

- Parent role with multiple student management
- Real-time chat (SignalR)
- SMS notifications
- Advanced search with Elasticsearch
- Tutor earnings dashboard
- Reviews and ratings system

9.3 Phase 3 Requirements

- Two-factor authentication
 - Mobile app (iOS/Android)
 - Video call integration
 - AI-powered tutor recommendations
 - Subscription plans for frequent users
-

10. Assumptions & Constraints

10.1 Assumptions

- Users have stable internet connection
- Users have valid email address
- Tutors have valid government ID
- Payment methods are credit/debit cards
- All content is in English initially

10.2 Constraints

- Development time: 30 days (2 hours/day)
 - Team size: 1 developer
 - Budget: Limited to free tiers of external services
 - Initial launch: Web only
-

11. Approval

Role	Name	Signature	Date
Product Owner	Muhammad Talha	Talha	20-04-26
Technical Lead	Muhammad Talha Muhammad Haseeb	Talha Haseeb	20-04-26
Developer	Muhammad Talha Muhammad Haseeb	Talha Haseeb	20-04-26
